

# *MATH V,* *(Optimization Theory: LPP)*

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# Introduction to Linear Programming Problems (LPP)

- Linear programming deals with the optimization (maximization or minimization) of a function of variables known as objective function, subject to a set of linear equalities and/or inequalities known as constraints.
- The objective function may be profit, cost, production capacity or any other measure of effectiveness, which is to be obtained in the best possible or optimal manner.
- The constraints may be imposed by different sources such as market demand, production process and equipment, storage capacity, raw material availability, etc. By linearity is meant a mathematical expression in which the variables do not have powers.
- It was in 1947 that George Dantzig and his associates found out a technique for solving military planning problems while they were working on a project for U.S. Air Force.
- This technique consisted of representing the various activities of an organization as a linear programming (PL) model and arriving at the optimal program by minimizing a linear objective function. He also developed the most powerful mathematical tool known as “Simplex method” to solve linear programming problems.

# Formulation of LPP

- Three fundamental concerns in formulating LP models are:
  - ① **Decision variables:** variables represent the decisions to be taken.
  - ② **Constraints:** What limits the decisions?
    - Main constraints: the restrictions and interactions on decision variable values.
    - Variable-type constraints: the set of values for which the variables have meaning
  - ③ **Objective function** : the function of the decision variables to be maximized or minimized.
- The General LPP can be written in the following form:

$$\text{Max}(\text{Min}) \quad z = \sum_{j=1}^n C_j X_j \quad (1)$$

$$\text{s.t} \quad \sum_{j=1}^n a_{ij} X_j \leq (\geq) b_i, \quad i = 1, \dots, m \quad (2)$$

$$X_j \geq 0, \quad j = 1, \dots, n \quad (3)$$

# Formulation of LPP: Application

## Example

**Production Allocation Problem:** A firm produces three products. These products are processed on three different machines. The time required to manufacture one unit of each of the three products and the daily capacity of the three machine are given in the table below.

| Machine | Time per unit (minutes) |           |           | Machine capacity<br>( minutes/day) |
|---------|-------------------------|-----------|-----------|------------------------------------|
|         | Product 1               | Product 2 | Product 3 |                                    |
| M1      | 2                       | 3         | 2         | 440                                |
| M2      | 4                       | -         | 3         | 470                                |
| M3      | 2                       | 5         | -         | 430                                |

It is required to determine the daily number of units to be manufactured for each product. The profit per unit for product 1, 2 and 3 is \$4, \$3 and \$6 respectively. It is assumed that all the amounts produced are consumed in the market.

# Formulation of LPP: Application

- **Step 1:** From the study of the situation find the key-decision to be made. In the given situation key decision is to decide the extent of product 1, 2 and 3, as the extents are permitted to vary.
- **Step 2:** Assume symbols for variable quantities noticed in step 1. Let the extents (amounts) of products, 1, 2 and 3 manufactured daily be  $x_1$ ,  $x_2$ , and  $x_3$  respectively.
- **Step 3:** Express the feasible alternatives mathematically in terms of variables. In the given situation feasible alternatives are sets of values of  $x_1$ ,  $x_2$ , and  $x_3$ , where  $x_1$ ,  $x_2$ , and  $x_3 \geq 0$ . Since negative production has no meaning and is not feasible.
- **Step 4:** Mention the objective quantitatively and express it as a linear function of variables. In the present situation, objective is to maximize the profit.

$$\text{Max } z = 4x_1 + 3x_2 + 6x_3 \quad (4)$$

# Formulation of LPP: Application

- **Step 5:** Put into words the influencing factors or constraints. These occur generally because of constraints on availability (resources) or requirements (demands). Express these constraints also as linear equalities/inequalities in terms of variables. Here, constraints are on the machine capacities and can be mathematically expressed as

$$2x_1 + 3x_2 + 2x_3 \leq 440$$

$$4x_1 + 0x_2 + 3x_3 \leq 470$$

$$2x_1 + 5x_2 + 0x_3 \leq 430$$

Therefore, the LPP is

$$\text{Max } z = 4x_1 + 3x_2 + 6x_3 \quad (5)$$

s.t

$$2x_1 + 3x_2 + 2x_3 \leq 440$$

$$4x_1 + 0x_2 + 3x_3 \leq 470$$

$$2x_1 + 5x_2 + 0x_3 \leq 430$$

$$x_1, x_2, x_3 \geq 0$$

# Formulation of LPP: Application

## Example

The WINDOR CO. produces high-quality glass products, It has three plants. Aluminum frames and hardware are made in Plant 1, wood frames are made in Plant 2, and Plant 3 produces the glass and assembles the products. Two new products are introduced and should be manufactured in the three plants. Product 1 is a glass door with aluminum frame and Product 2 is a wood-framed window. The production time(hour/per batch) and the profit(per batch) for each of the products, and the product time available in each plant are shown in the following table.

| Plant  | Production time |           | Production time available per week |
|--------|-----------------|-----------|------------------------------------|
|        | Product 1       | Product 2 |                                    |
| 1      | 1               | 0         | 4                                  |
| 2      | 0               | 2         | 12                                 |
| 3      | 3               | 2         | 18                                 |
| Profit | \$3,000         | \$5,000   |                                    |

Determine the production rates of the two products in order to maximize company's total profit.

# Formulation of LPP: Application

① Define the decision variables:

- $x_1$  = number of batches of product 1 produced per week
- $x_2$  = number of batches of product 2 produced per week

② Main constraints:

- In plant 1 no more than 4 hours per week may be used to produce product 1.  $\Leftrightarrow x_1 \leq 4$
- In plant 2 no more than 12 hours per week may be used to produce product 2.  $\Leftrightarrow 2x_2 \leq 12$
- In plant 3 no more than 18 hours per week may be used to produce products 1 and 2.  $\Leftrightarrow 3x_1 + 2x_2 \leq 18$

③ Variable-type constraints: production rates can not be negative.  $\Leftrightarrow x_1, x_2 \geq 0$

④ Objective function:

Let  $z$  = total profit in thousands of dollars gained by producing  $x_1$  and  $x_2$  batches of products 1 and 2.

The objective is to maximize the function is  $z = 3x_1 + 5x_2$ .



# Formulation of LPP: Application

The mathematical model of that WINDOR CO cited above can be written as the following:

$$\begin{aligned} \text{Max } z &= 3x_1 + 5x_2 \\ \text{s.t } x_1 &\leq 4 \\ 2x_2 &\leq 12 \\ 3x_1 + 2x_2 &\leq 18 \\ x_1, x_2 &\geq 0 \end{aligned}$$

This model is an LP problem.

# Solution to LPP

- An  $n$  – tuple  $(x_1, x_2, \dots, x_n)$  of real numbers which satisfies the constraints (2) and (3) of a General LPP given above is called a **solution to the general LPP**.
- Any solution to a general LPP which also satisfies the non-negative restrictions of the problem, is called a **feasible solution** to the general LPP
- Any feasible solution which optimizes ( minimizes or maximizes) the objective function (1) of a General LPP is called an **optimum solution** to the General LPP.

# Graphical solution method: Search Approach method

The major steps in the solution of linear programming problem by graphical method are summarized as follows:

- 1 Identify the problem-the decision variables, the objective and the restrictions.
- 2 Set up the mathematical formulation of the problem
- 3 Plot a graph representing all the constraints of the problem and identify the feasible region (solution space). The feasible region is the intersection of all the regions bounded ( represented) by the constraints of the problem and is restricted to the first quadrant only.
- 4 The feasible region obtained in step 3 may be bounded or unbounded
- 5 Find out the value of the objective function at each corner (solution) point determined in step 4
- 6 Select the corner point that optimizes (maximizes or minimizes) the value of the objective function. It gives the optimum feasible solution

# Graphical solution method: Iso-profit or Iso-cost approach

## Remark

- 1 The above method is known as **Search approach method**
- 2 Another method known as **Iso-profit or Iso-cost approach**, involves the following steps

The Iso-profit or Iso-cost approach, involves the following steps:

- a First four steps are same as in the Search Approach method. In the fifth step we choose a convenient profit (or cost) and draw iso-profit (iso-cost) line so that it falls within the feasible region.
- b Move this iso-profit (iso-cost) line parallel to itself farther (closer) from (to) the origin.
- c Identify the optimum solution as the coordinates of that point on the feasible region touched by the highest possible iso-profit line ( or lower possible iso-cost line)
- d Compute the optimum feasible solution.

# Graphical solution

- Let us solve graphically the mathematical model (6) rewritten as:

$$\text{Max } z = 3x_1 + 5x_2$$

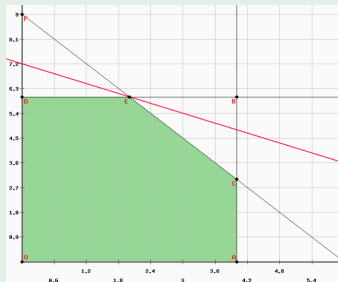
$$\text{s.t. } x_1 \leq 4$$

$$2x_2 \leq 12$$

$$3x_1 + 2x_2 \leq 18$$

$$x_1, x_2 \geq 0$$

- The region bounded by the constraints of the above model (6) is



# Graphical solution

- The optimal solution is  $z = 36$  units, occurred at the point  $E(2, 6)$  as shown in this table below.

| Point | X coordinate ( $X_1$ ) | Y coordinate ( $X_2$ ) | Value of the objective function ( $Z$ ) |
|-------|------------------------|------------------------|-----------------------------------------|
| O     | 0                      | 0                      | 0                                       |
| A     | 4                      | 0                      | 12                                      |
| B     | 4                      | 6                      | 42                                      |
| C     | 4                      | 3                      | 27                                      |
| D     | 0                      | 6                      | 30                                      |
| E     | 2                      | 6                      | 36                                      |
| F     | 0                      | 9                      | 45                                      |
| G     | 6                      | 0                      | 18                                      |

# Exercises

- ① Find the optimal solution of the following LP problem:

$$\begin{aligned} \text{Max } z &= 4x + 3y \\ \text{s.t. } & -x + y \leq 14 \\ & 2x + y \leq 16 \\ & x, y \geq 0 \end{aligned}$$

- ② A factory manufactures nails and screws. The profit yield is \$2 per kg nails and \$3 per kg screws. Three units of labour are required to manufacture 1 kg nails and 6 units to make 1 kg screws. Twenty-four units of labour are available. Two units of raw material are needed to make 1 kg nails and 1 unit for 1 kg screws. Determine the manufacturing policy that yields maximum profit from 10 units of raw material.
- ③ Sketch the constraints.

$$\begin{aligned} 2x - y &\leq 6 \\ x + 2y &\leq 8 \\ 3x + 2y &\leq 18 \\ y &\leq 3 \end{aligned}$$

and verify that the maximum of the function  $x + y$  in the feasible region is at  $x = 4$  and  $y = 2$ .

# Introduction

- Simplex method, also called **simplex technique** or **simplex algorithm** was developed by G.B. Dantzig, an American Mathematician. It has the advantage of being universal, i.e., any linear model for which the solution exists, can be solved by it. In principle, it consists of starting with a certain solution of which all that we know is that it is feasible, i.e., it satisfies the non-negativity conditions ( $x_j \geq 0, j = 1, 2, \dots, n$ ).
- We then, improve upon this solution at consecutive stages, until, after a certain number of stages, we arrive at the optimal solution.
- The basis of the simplex method consists of two fundamental conditions:
  - 1 The **feasible condition**: It ensures that if the starting solution is basic feasible, only basic feasible solutions will be obtained during computations.
  - 2 The **optimal condition**: It guarantees that only better solutions (as compared to the current solution) will be encountered.
- The **simplex method** is an iterative algorithm that begins with an initial feasible solution, repeatedly moves to a better solution, and stops when an optimal solution has been found and, and then no improvement can be made.



# Canonical Form

- The general LPP discussed in previous section can always be put in the following form called the **canonical form**:

$$\begin{aligned} \text{Max } z &= \sum_{j=1}^n C_j X_j \\ \text{s.t } \sum_{j=1}^n a_{ij} X_j &\leq, b_i, & i = 1, \dots, m \\ X_j &\geq 0, & j = 1, \dots, n \end{aligned}$$

- The characteristics of the canonical form are:

- 1 All decision variables are non-negative,
- 2 All constraints are of the ( $\leq$ ) and
- 3 Objective function is of maximization type.

# Standard form

Let us consider the general linear programming problem

$$\begin{aligned} \text{Max } z &= \sum_{j=1}^n C_j x_j \\ \text{s.t } \sum_{j=1}^n a_{ij} x_j &\leq b_i, (\geq b_i) & i = 1, \dots, m \\ x_j &\geq 0, & j = 1, \dots, n \end{aligned}$$

This expressed in the standard form as

$$\begin{aligned} \text{Max } z &= \sum_{j=1}^n C_j x_j + 0s_1 + \dots + 0s_m \\ \sum_{j=1}^n a_{ij} x_j + s_i &= b_i, & i = 1, \dots, m \\ x_j &\geq 0, & j = 1, \dots, n \\ s_i &\geq 0, & i = 1, \dots, m \end{aligned}$$

# Matrix form of LPP

- Consider the standard form of an LP in matrix form:

$$\begin{aligned} \text{Max } z &= c^T x \\ \text{s.t } Ax &= b \\ x &\geq 0 \end{aligned}$$

## Definition

Consider a system of linear equations  $Ax = b$  with  $m$  equations and  $n$  variables (assume  $n \geq m$ ). A **basic solution** to the system of linear equations  $Ax = b$  is obtained by setting  $n-m$  variables equal to 0 and solving for the values of the remaining  $m$  variables. Then,

- Any basic solution to the LP problem in which all variables are nonnegative is called a **basic feasible solution**. Otherwise, the basic solution is infeasible.
- The variables that are set to zero are called **nonbasic variables**
- The remaining ones are called **basic variables**
- The columns corresponding to the basic variables are linearly independent.

# Algorithm of Simplex Method

- **Step 0: Initialization**

Convert the LP to standard form and find an initial basic feasible solution  $x^{(0)}$ .  
Set  $k = 0$

- **Step 1: Optimality test**

If the coefficients of all nonbasic variables in the objective function row (denoted by  $c_j, \forall j$ ) satisfy the following

- 1  $c_j \geq 0, \forall j$  (for maximization problem)
- 2  $c_j \leq 0, \forall j$  (for minimization problem)

then the basic feasible solution  $x^{(k)}$  is optimal. Otherwise, go to **Step 2**.

**Note:** The LP problem has multiple optimal solutions if, at least, one of the nonbasic variable has a zero coefficient in the optimal tableau's objective function row.

# Algorithm of Simplex Method

- **Step 2:** Determine the entering variable Choose the nonbasic variable  $x_p$  that has

- 1 For maximization problem

$$c_p = \min_j \{c_j | c_j < 0\}$$

- 2 For minimization problem

$$c_p = \min_j \{c_j | c_j > 0\}$$

to enter the basis.

- **Step 3:** Determine the leaving variable Compute the ratio

$$\frac{b_i}{a_{ij}} = \min_i \left\{ \frac{b_i}{a_{ij}} | a_{ij} > 0 \right\}$$

where  $b_i$  is the right-hand-side coefficient of equation  $i$  and  $a_{ij}$  is the coefficient of entering basic variable  $x_j$  in row  $i$ . Then the basic variable in row  $i$  is the leaving basic variable.

**Note:** If the coefficients  $a_{ij} < 0$  for all  $i$ , then the LP problem is **unbounded**.

- **Step 4:** Determine the new basic feasible solution  $x^{(k+1)}$  by **Gaussian Elimination**. Update  $k := k + 1$  and go to **Step 1**.

# Algorithm of Simplex Method

## Example

Use simplex method to solve the following Linear Programming Problems:

1

$$\text{Max } z = 8x_1 + 5x_2$$

$$\text{s.t. } x_1 \leq 150$$

$$x_2 \leq 250$$

$$2x_1 + x_2 \leq 500$$

$$x_1, x_2 \geq 0$$

2

$$\text{Max } z = 30x_1 + 20x_2$$

$$\text{s.t. } 2x_1 + x_2 \leq 150$$

$$2x_1 + x_2 \leq 100$$

$$x_1 + x_2 \leq 80$$

$$x_1 \leq 40$$

$$x_1, x_2 \geq 0$$